

colliding packets in WSNs. Some others avoid unnecessary reception of packets by nodes that are unintended destinations. It has been assessed that idle power consumption can be of the same order as that consumed by the transmitter and receiver. Some of the MAC protocols are described below.

S-MAC. This protocol creates a sleep schedule that determines when to activate the receivers and when to put them in sleep mode.

Timeout-MAC (T-MAC). It eliminates idle energy further; instead of allowing the messages to be sent continuously, messages are transmitted in bursts in the beginning of the frame.

DMAC. Since many WSNs have data-gathering trees routed to a single data sink in the direction of packets arriving at a node, DMAC takes advantage of this by staggering the wakeup times for nodes based on their distance from the data sink.

Traffic-adaptive medium access (TRAMA). Aforementioned protocols help in minimising the power consumption by reducing the time that the transmitters remain in idle state, TRAMA attempts to reduce wasted energy consumption caused by packet collisions. Nodes determine their transmitting state using adaptive election algorithm (AEA). In AEA, each node calculates a priority for itself and all two-hop neighbours for the current slot. If a node has the highest priority for that slot and has data to send, it wins that slot and sends the data. If one of its neighbours has the highest priority, it sets itself to the receiving mode. In short, AEP assigns priorities for the unused slots to the nodes needing extra slots.

Sparse topology and energy management (STEM). When data packets are generated, the sensor generating the traffic uses a paging channel (separate from the data channel) to awaken the downstream neighbours.

Storage technology for portable devices

Flash memory has been the most reliable storage technology. However, visible limitations for future Flash cell

scaling include power consumption, charge storage requirements of the dielectrics, reliability issues and capacitive coupling between adjacent cells. Efforts are being made to reduce these limitations through system management techniques and fabrication technology (such as high k-dielectrics, nano crystal storage media and FinFET). However, there is no doubt that Flash will remain the dominant non-volatile memory technology at least down to 45nm node. Nevertheless, there are new memory technologies available, such as ferro-electric RAM (FeRAM), magnetic RAM (MRAM), ferro-electric polymer RAM (FePRAM), phase change memory (PCM), resistive RAM (RRAM), probe storage, carbon nano-tube memory (CNT) and molecular memory. The most promising are probe storage memories and PCM.

IEEE 802.16

IEEE 802.16 working group has come up with a standard called broadband wireless access (BWA), commonly known as WiMAX (worldwide interoperability for microwave access). IEEE 802.16 has specified two aspects of the interface for WiMAX, the physical layer (PHY) and the media access control layer (MAC).

The physical layer defines electrical and physical specifications for a device's establishment and termination of connection to communication medium, communication flow control, modulation coding, etc. It was designed to accommodate the need for a low cost yet allowing for high levels of integration. The use of direct sequence allows the analogue circuitry to be very simple and very tolerant towards inexpensive implementations.

The MAC layer was designed to allow multiple topologies without complexity. The power management operation does not require multiple modes of operation. The MAC allows reduced functionality device (RFD) that need not have Flash or large amounts of ROM or RAM.

MAC is divided into three sub-layers: convergence, common part and

security (or privacy). The convergence sub-layer describes how wireline technologies, such as asynchronous transfer mode (ATM), Ethernet 802.1 (LAN/MAN) and Internet protocol (IP) are encapsulated on the air interface and how data is classified.

The common part sub-layer is responsible for idle-mode processes like cell selection, paging structures and location-area updates. This layer is also responsible for sleep-mode processes, handover procedures, multicast and broadcast services, quality-of-services class and automatic repeat request (ARQ) processes. It also does header suppression, packing and fragmentation for efficient use of spectrum.

The security sub-layer provides the subscribers with privacy, authentication or confidentiality across the broadband wireless network. It uses encapsulation protocol and privacy key management (PKM) protocol. One of the features of MAC layer of 802.16 is that it is designed to differentiate service among traffic categories with different multimedia.

WiMAX is a wireless metropolitan area network technology which fits between wireless local area networks and wireless wide area networks. This standard introduces many advantages including variable and high data rate (up to 75Mbps), last-mile wireless access (up to 50km), point-to-multipoint communication, large frequency range and quality of service.

WiMAX supports two network architectures, namely, point-to-multipoint (PMP) mode and mesh mode. PMP network has one base station (BS) and multiple subscriber stations (SSs). The BS is the central control point and regulates all the traffic on the network. The multiple access schemes in WiMAX include both frequency division duplexing (FDD) and time division duplexing (TDD). TDD is more flexible than FDD.

The performance of IEEE 802.16 is assessed by throughput and delay. Throughput of scheduler is defined as the maximum bandwidth allocated by the scheduler with respect to the requested data rate by the subscriber stations, and the delay is defined as the